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Curing Kinetics and Structure-Property Relationship of Moisture-Cured One-Component Polyurethane Adhesives

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ABSTRACT

Moisture-curable polyurethane (PU) adhesives have been exploited in the automotive industry because of their usability, non-toxicity without the use of volatile organic compounds (VOC), outstanding solvent resistance, and adhesiveness. However, there has been a lack of quantitative and comprehensive analyses related to the chemical and thermomechanical properties of moisture-curable PU adhesives depending on the applied curing conditions. Herein, we systematically investigated the effects of curing conditions, including relative humidities (25, 35, 45, 55, 65, and 75%RH) and curing duration (0, 0.25, 0.5, 1, 2, 3, 4, 5, 6, and 7 days), on the structure–property relationships of moisture-cured PU adhesives. As the relative humidity increased, the chemical reaction of the PU adhesives accelerated, and they exhibited a high crosslinking density and storage modulus. However, even prolonged curing time and higher relative humidity did not lead to a continuous enhancement in the mechanical properties of the PU adhesive. After a sufficient curing time, the storage modulus of the PU adhesive reached saturation, and the PU adhesives cured above 65%RH demonstrated a comparable saturated storage modulus. This structure–property relationship of PU adhesives will provide fundamental understanding and insights for quality control and optimization of processing time in the industry.

1. Introduction

Polyurethane (PU) is widely used in various industrial fields because of its versatile mechanical properties resulting from the diverse combinations of isocyanates (–NCO) and polyols. Typically, PU exhibits segmented structures based on alternately interconnected hard and soft segments [1]. Aliphatic polyols act as flexible long chains with high segmental mobility, which form the soft segments. Urethane bonds contribute to the formation of hard segments by the molecular assembly of urethane bonds via strong intermolecular hydrogen bonding [2,3,4]. Although high-molecular-weight polymers entropically prefer random coil structures, the enthalpic contribution of hydrogen bonding in urethane groups can compensate for this entropic penalty, leading to dense molecular packing with positional order in structurally hard crystals [5,6]. The hard segment tends to aggregate and form a hard domain because of hydrogen bonding and thermodynamic incompatibility with the soft segment [7]. An increase in the number of hard domains enhanced rigidity, whereas a decrease in the number of hard domains increased flexibility. Hence, different combinations of –NCO and polyols can provide a wide range of mechanical properties for PU, from soft to rigid, by tailoring the composition of the hard domain [1]. This versatile PU has been widely applied in diverse fields such as adhesives [8,9,10,11,12,13,14], foams [15,16], elastomers [17,18,19], coatings [20], and nanocomposites [21,22]. Furthermore, compared to typical polymers synthesized through condensation reactions, the molecular weight of PU can be easily controlled by adjusting the number of reactants, including –NCO and polyols, owing to the absence of byproducts (e.g., water) during the synthesis of polyurethane. This implies that the chemical equilibrium shifts towards the product without any interference, according to Le Chatelier's principle [23,24].

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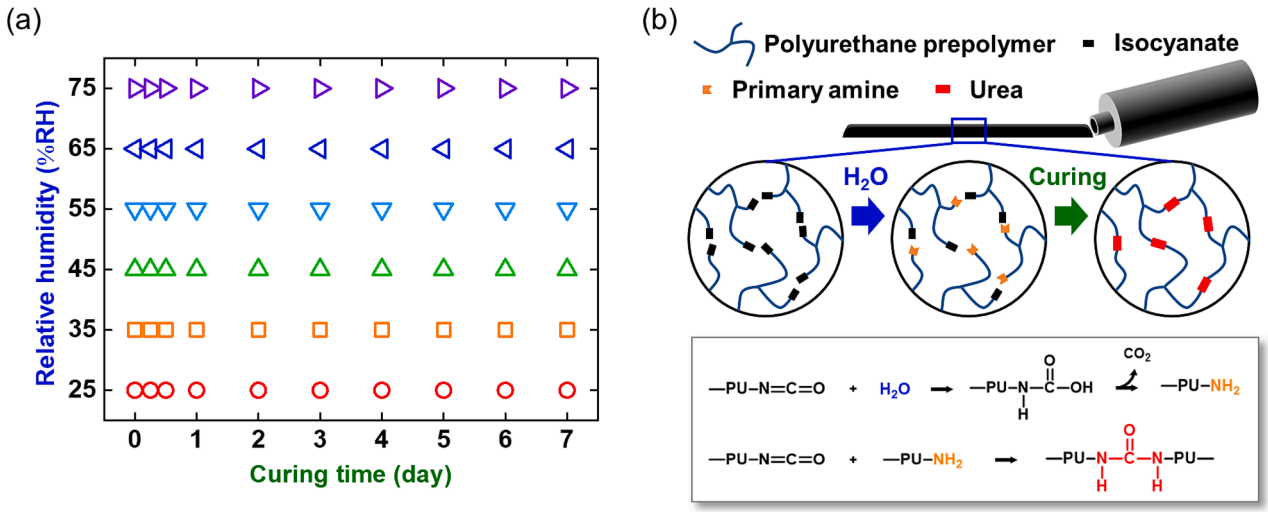

Fig. 1. Evaluation for curing behavior of PU adhesive depending on the curing conditions. (a) Measurement curing conditions for evaluating the properties of PU adhesive as a function of curing conditions including curing time and relative humidities. (b) Schematic illustrations for moisture curing of PU adhesive and chemical reaction on moisture curing of –NCO with moisture.

In particular, moisture-curable PU adhesives have been widely used in the automotive industry as one-component adhesives capable of undergoing curing reactions with moisture in air [25,26]. Compared to two-component adhesives that require the physical mixing of a base and a hardener, moisture-curable PU adhesives offer more convenient curing options through the reaction with atmospheric moisture under ambient temperature and pressure conditions [27,28]. Typically, a moisture-curable adhesive composed of a plasticizer that enhances the adhesion strength and flexibility [29], fillers such as carbon black and calcium carbonate ($CaCO_3$) used to adjust the adhesive properties [30], and a PU prepolymer containing –NCO functional groups are cured by forming urea bonds through the reaction of –NCO with water molecules. First, the –NCO functionalized in the PU prepolymer reacts with moisture in ambient air, resulting in the formation of primary amines and the release of CO_2 [8,31]. Subsequently, these primary amines react with other –NCO groups to form urea bonds, ultimately leading to the formation of a urea-urethane copolymer [32,33]. The moisture-dependency of these reactions is influenced by the adhesive surface moisture and the relative humidity of the surrounding air, which plays a crucial role in the curing process and affects the chemical conversion rate, degree of conversion, molecular structure, and mechanical properties of the final product [11,12,13]. Furthermore, even under the same relative humidity, the degree of crosslinking can vary over time, making it essential to understand the kinetics of crosslinking reactions in relation to curing conditions, including relative humidity and curing time. Nevertheless, few comprehensive and systematic evaluations of the interconnected chemical, mechanical, and rheological properties have been conducted on the structure–property relationship under various curing conditions [27,34].

In this study, we systematically investigated humidity-dependent curing kinetics of the one-component moisture curable PU adhesive named direct glazing urethane (DGU) by varying relative humidities (25, 35, 45, 55, 65, and 75%RH) and curing duration (0, 0.25, 0.5, 1, 2, 3, 4, 5, 6, and 7 days) at an identical temperature of 20 °C. A thermo-hygrostat chamber was employed to maintain consistent temperature and relative humidity during the curing process. The structure–property relationship based on the curing kinetics is defined by considering the rheology, Fourier-transform infrared (FT-IR) spectra, thermal degradation behavior, and thermomechanical properties. The viscoelastic properties of the uncured PU adhesive in paste form were measured using a rheometer, whereas the elastic modulus of the cured PU adhesive that retained its shape was measured using a dynamic mechanical analyzer (DMA). Additionally, the variation in intensity of the –NCO peak was evaluated using FT-IR spectroscopy, and the different thermal degradation behaviors of the chemical components were evaluated using thermogravimetric analysis (TGA) [35]. Based on these structure–property relationships, we discuss the influence of curing conditions, including relative humidity and curing time, on the curing kinetics and mechanical properties of the final product, suggesting the significance of such insights in both academia and industry.

2. Experimental

2.1. Preparation of specimen for analysis

The moisture-curable PU adhesive employed in this work was direct glazing urethane (DGU; PU9510, KCC Corporation) provided by the Hyundai Motor Company. 3D printed negative molds were fabricated using stereolithography apparatus (SLA) technology with a polylactic acid (PLA)-based photocurable resin (eResin PLA BIO-Photopolymer Resin, eSUN). The PU adhesive was filled into two types of negative molds to prepare the specimens for the TGA and DMA tests. The diameter and height of the cylinder in the negative mold for TGA were 8 and 3 mm, respectively (Fig. S1a). The width, length, and thickness of the DMA specimens were 2, 50, and 1 mm, respectively (Fig. S1b). For the FT-IR specimens, 3 mg of the PU adhesive was applied between two KBr pellets with a diameter of 13 mm. To avoid the effects by sample amount, FT-IR analysis was carried out depending on curing time using a single PU adhesive for each relative humidity condition. The PU adhesive was applied between two steel substrates according to ASTM D906 for the universal testing machine (UTM) specimens. The PU adhesive was then cured in a thermo-hygrostat chamber under different relative humidities of 25, 35, 45, 55, 65, and 75%RH and a constant temperature of 20 °C for 0.25, 0.5, 1, 2, 3, 4, 5, 6, and 7 days.

2.2. Fourier transform-infrared spectroscopy

The chemical bonding characteristics of the PU adhesive were evaluated using an FT-IR spectrometer (Nicolet iS5, Thermo Fisher Scientific) in the wavenumber range of 4000–800 cm^{-1} at a resolution of 4 cm^{-1}.

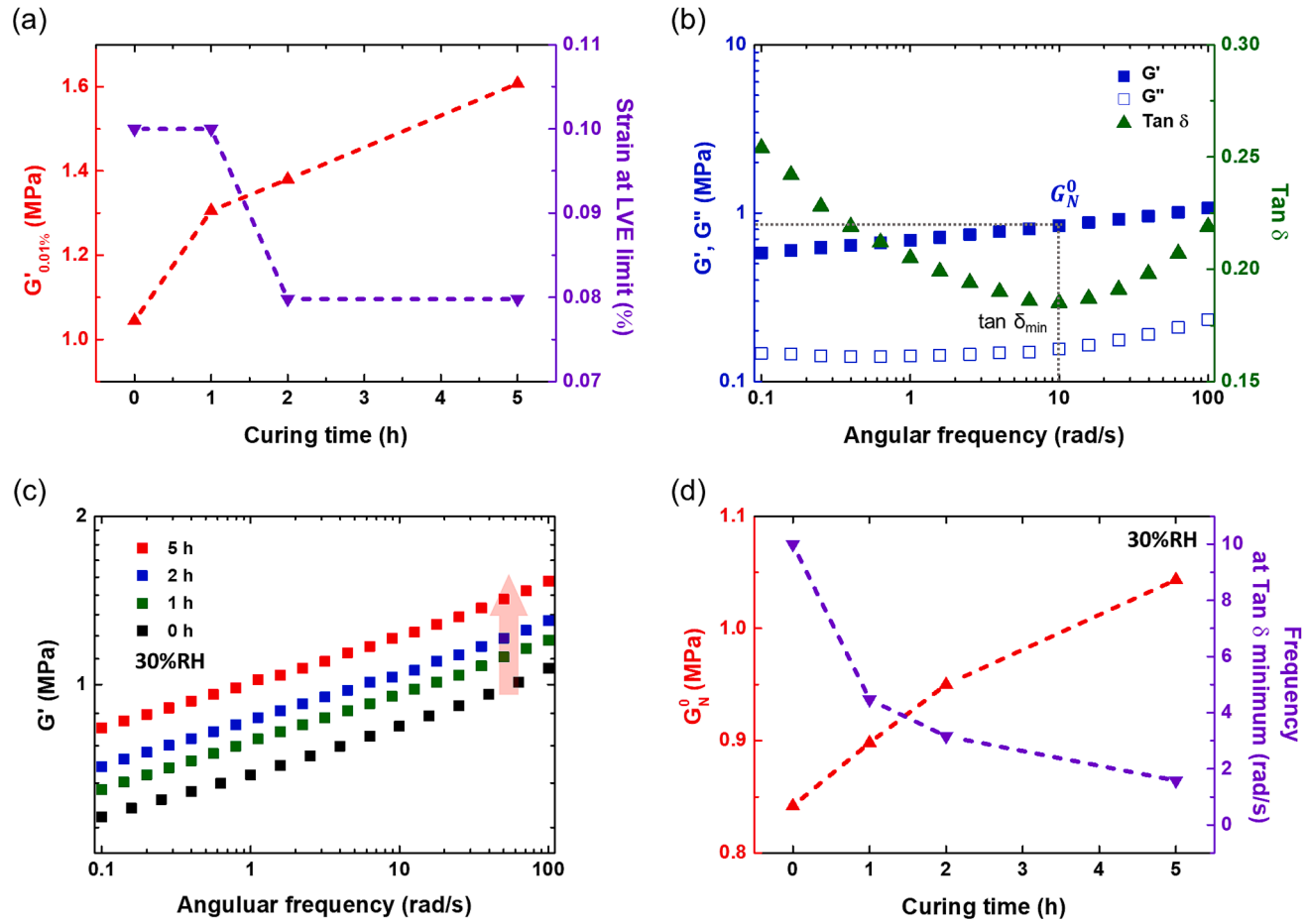

Fig. 2. Rheological properties of PU adhesive. (a) Storage modulus at strain of 0.01% ($G'_{0.01\%}$) and strain at LVE limit of PU adhesive as a function of curing time. (b) Storage modulus (G'), loss modulus (G'') and loss tangent (tan δ) of uncured PU adhesive as a function of angular frequency. (c) Storage modulus of the PU adhesive depending on curing time as a function of angular frequency. (d) Plateau modulus (G_N^0) and frequency at Tan δ minimum of the PU adhesive as a function of curing time.

2.3. Mechanical properties

The rheological properties of the PU adhesive were evaluated by a strain-controlled rheometer (MCR 302e, Anton Par) at 20 °C. The linear viscoelastic (LVE) region of the PU adhesive was determined using a strain sweep test at a frequency of 100 rad/s. Subsequently, a small-amplitude oscillatory shear (SAOS) test was conducted to define the rheological properties of the PU adhesive cured under ambient conditions of 30%RH and 20 °C for 0, 1, 2, and 5 h. The frequency sweep tests were conducted within a frequency range of 0.1 to 100 rad/s at a strain of 0.05%.

2.4. Thermal properties

The thermal degradation behavior of the PU adhesives was evaluated using TGA (Q50, TA Instruments) as a function of the curing time. TGA thermograms of the PU adhesives (50 mg) were recorded while heating from room temperature to 900 °C at a heating rate of 20 °C/min under a nitrogen atmosphere. Variations in the chemical composition of the PU adhesive were identified by evaluating the derivative thermogravimetry (DTG) curves obtained from the TGA thermograms.

2.5. Thermomechanical properties

Thermomechanical properties of the PU adhesive were measured by a stress-controlled DMA (Q800, TA Instruments) over the temperature range from 25 to 120 °C with a frequency of 1 Hz and amplitude of 10 μm with a heating rate of 5 °C/min. The gauge length, width, and thickness of the specimens were 10, 3, and 1 mm, respectively.

2.6. Adhesion properties

Adhesion properties of the PU adhesive were evaluated by lap shear test using a UTM (5966, Instron) equipped with a 10 kN cell operating at 1.0 mm/min crosshead speed. The length, width, and thickness of applied PU adhesive were 2, 2.5, and 0.3 mm, respectively.

3. Results and discussion

To comprehensively evaluate the effects of the curing conditions on the reaction kinetics and crosslinked structure of PU adhesives, we systematically cured the DGU under different curing conditions (Fig. 1a). Typical adhesives are formulated in paste form with a specific viscosity to prevent undesired flow and maintain their shape when applied to the desired areas. PU adhesives are prepolymers with reactive urethane bonds containing –NCO groups capable of reacting with water and are cured through a chemical reaction of –NCO with water molecules (Fig. 1b). To evaluate the rheological behavior of the PU adhesive in the paste state, in which its shape can be altered by external stress, measurements were conducted using a rheometer. The initial viscoelastic response to curing in the PU adhesive was evaluated within the LVE region for curing times of 0, 1, 2, and 5 h. The maximum strain of the PU adhesive within the LVE region was determined from the results of strain sweep tests (Fig. 2a). As the PU adhesive cured over time, the

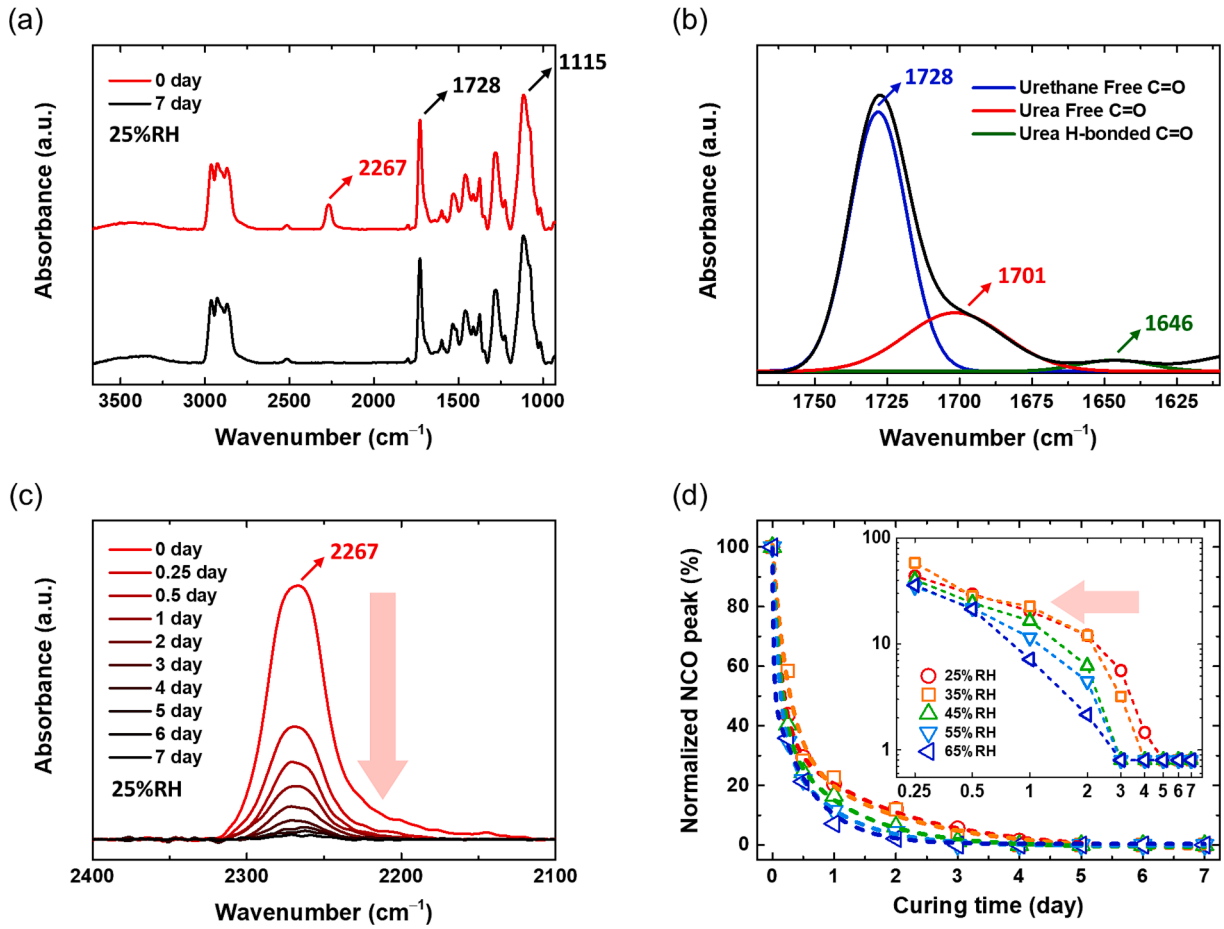

Fig. 3. FT-IR absorption spectra of PU adhesive. (a) Overall spectra of uncured PU adhesive and PU adhesive cured at 25%RH and 20 °C for 7 days. (b) Deconvolution curve of C = O peak region for urethane and urea. (c) –NCO peak (2267 cm^{-1}) depending on curing time at 25%RH and 20 °C. (d) Normalized –NCO peak of PU adhesive cured at different relative humidity and curing time at a constant temperature of 20 °C. The dashed lines indicate fitting via two-phase exponential decay as a function of curing time.

segmental mobility of the internal chains was restricted, resulting in a reduction in the maximum strain within the LVE region from 0.1 to 0.08% (Fig. S2). Additionally, the PU adhesive exhibited an increased storage modulus from 1.05 MPa for the uncured adhesive at an initial strain of 0.01% to 1.60 MPa when cured for 5 h.

The plateau modulus (G_N^0) of the PU adhesive was defined as the storage modulus at the angular frequency where the loss tangent was minimum (Fig. 2b and S3) [36]. PU adhesives possess cross-linked bonds that form a network structure, resulting in a plateau region. However, the plateau region of the PU adhesive exhibits a slope because of the use of a prepolymer with a small molecular weight and the addition of plasticizers. The storage modulus of the PU adhesive at an angular frequency of 0.1–100 rad/s gradually increased with curing time at 30%RH (Fig. 2c). As the curing time progressed for 5 h at 30%RH, the plateau modulus of the initial PU adhesive increased from 0.84 to 1.04 MPa, while the frequency when the loss tangent is minimum decreased from 10 to 1.58 rad/s (Fig. 2d). These results imply that the curing degree of the PU adhesive increases with exposure to moisture. The uncured PU adhesive paste exhibited an increase in the elastic behavior depending on the curing time.

The conversion of –NCO and chemical structures of the PU adhesives were evaluated by measuring the transmission mode of the FT-IR spectroscopy. Fig. 3a depicts the FT-IR spectra of the uncured PU adhesive in its initial prepolymer and the PU adhesive cured at 25%RH and 25 °C for 7 days. The range of 2800–3000 cm^{-1} was assigned to the stretching vibrations of both the antisymmetric and symmetric stretching of the methyl and methylene (C–H) bonds [37]. The peak at 2267 cm^{-1} indicates the presence of free –NCO in the terminal functional group of the PU adhesive [37,38]. The spectrum of the uncured PU adhesive exhibited a prominent –NCO peak, which eventually disappeared after complete curing. The peak at 1115 cm^{-1}, observed for both uncured and cured PU adhesives, was attributed to the characteristic ether (C-O-C) group in the polyol, which contributes to the formation of the soft segment within the polyurethane [37]. The sharp peak at 1726 cm^{-1} indicates the stretching mode of the carbonyl (C = O) bond in both the uncured and cured PU adhesives. This C = O peak observed in the spectral range of 1625–1750 cm^{-1} for uncured PU adhesive was deconvoluted using a curve fitting method by employing a Gaussian function (Fig. 3b). The three different overlapped peaks in 1728, 1701, and 1646 cm^{-1} were obtained and corresponded to the free C = O in urethane, urea, and H-bonded C = O in urea, respectively [39]. These results confirm the presence of both urethane and urea bonds in the PU adhesive prepolymer. The H-bonded C = O bonds of urea contributed to strong intermolecular hydrogen bonding. Additionally, the peaks observed at 871, 1406, and 3641 cm^{-1} in the PU adhesive residue after thermal decomposition indicated the characteristic peak of calcium oxide (CaO) generated by the thermal decomposition of $CaCO_3$ (Fig. S4a) [40].

As shown in Fig. 3c, the absorbance of the –NCO peak progressively decreased as the PU adhesive was cured. Upon reacting with the ambient moisture, the terminal –NCO groups react with H_2O molecules over time, resulting in the absence of a peak near 2267 cm^{-1} for PU adhesive cured for 7 days. These results suggest that moisture curing leads to the formation of urea bonds via the chemical reaction of the –NCO groups

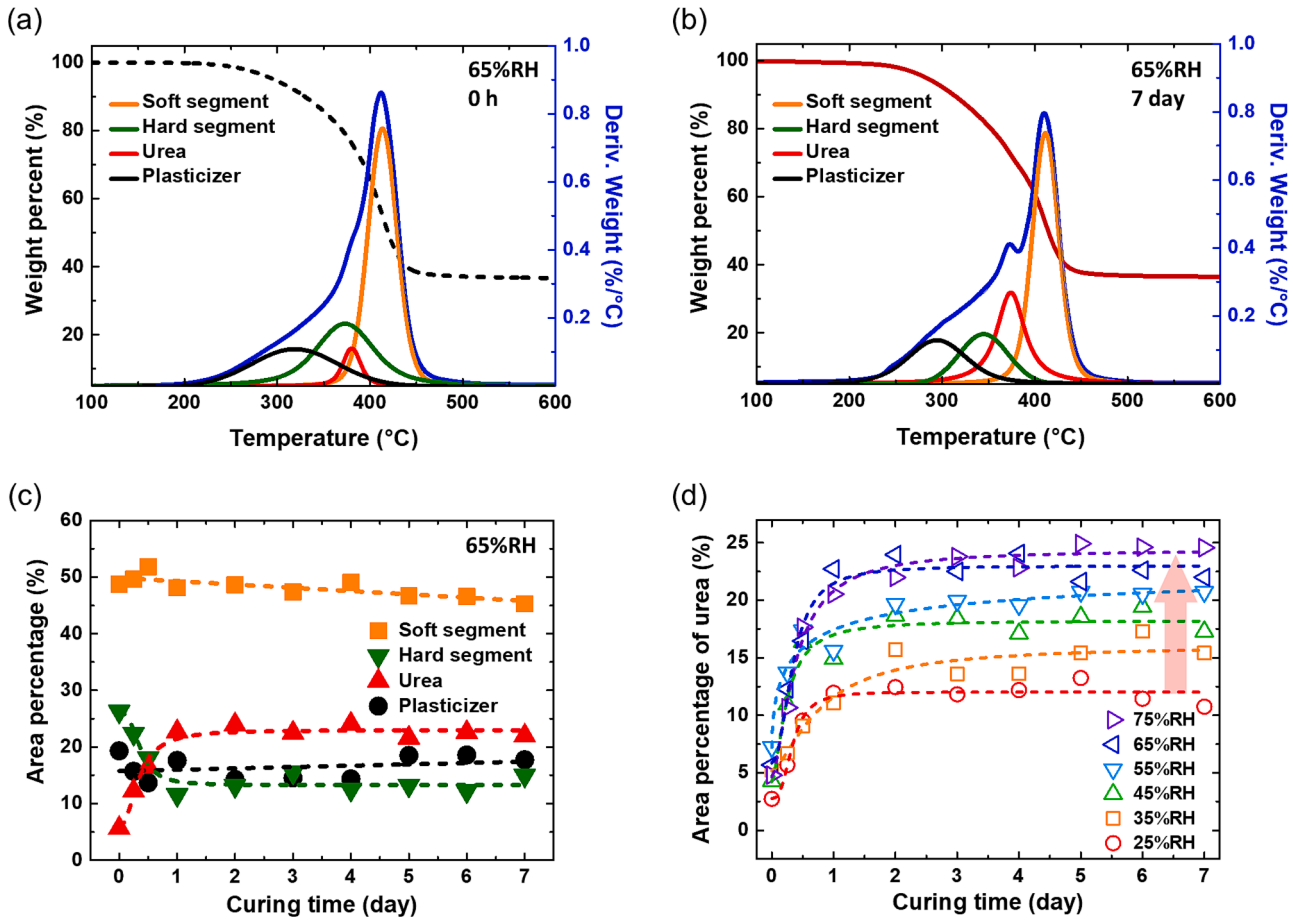

Fig. 4. Thermal decomposition analysis of PU adhesive. TGA thermograms, DTG curves, and deconvolution of the DTG curves of (a) uncured PU adhesive, (b) PU adhesive cured at 65%RH and 20 °C for 7 days. (c) Area percentage of chemical composition within the PU adhesive cured at 65%RH and 20 °C reproduced by deconvolution of DTG curves depending on curing time. (d) Area percentage of urea within the PU adhesive cured at different relative humidity and curing time at a constant temperature of 20 °C. The dashed lines indicate fitting via hill equation as a function of curing time.

with water molecules. To evaluate the conversion of the moisture-driven curing process caused by the consumption of –NCO, the absorbance was normalized by calculating the ratio of the –NCO absorption peak to that of the uncured PU adhesive. Fig. 3d shows the evolution of the normalized –NCO peak intensity as a function of curing time under different relative humidity conditions. The normalized –NCO peaks exhibited an exponential decay with increasing curing rate and relative humidity. At 25%RH, 5 days were required for the -NCO peak to disappear, whereas the –NCO peak disappeared after 4 days at 35%RH. In the case of the PU adhesive cured at 45, 55, and 65%RH, the –NCO peak disappeared after 3 days of curing. Abundant moisture promotes the chemical reaction of the PU adhesive terminated with the –NCO group owing to an increase in the number of reaction sites, which accelerates the curing of the PU adhesive.

We evaluated the thermal stability and variation in the chemical composition by analyzing the TGA thermograms and the corresponding DTG curves of the PU adhesive. Polyurethane undergoes a two-step decomposition process in which the hard segment first decomposes, followed by the subsequent decomposition of the soft segment [41]. Furthermore, the decomposition of the urethane bonds in the hard segment occurred more rapidly than that of the urea bonds [42]. Therefore, by analyzing the decomposition behavior of urea and urethane bonds through TGA, variations in the composition depending on the curing conditions could be observed. Fig. 4a and 4b show the TGA thermogram and DTG curve in the organic decomposition region less than 500 °C of the uncured and cured PU adhesives at 65%RH and 20 °C for 7 days, respectively. The degradation temperature at 5% weight loss ($T_{d,\ 5\ wt\%}$) of the uncured PU adhesive was measured as 297 °C, whereas $T_{d,\ 5\ wt\%}$ of the PU adhesive cured for 7 days is measured as 282 °C, indicating faster degradation as curing proceeded. The decrease in thermal degradation temperature implies that the plasticizers confined between the segments tend to migrate closer to the surface, leading to accelerated decomposition. This migration is facilitated by the aggregation of segments, which occurs because of the increased hydrogen bonding resulting from the formation of urea bonds. Within the DTG curve of the PU adhesive, multiple Voigt functions were convoluted based on the decomposition temperature of the chemical structure including plasticizer, urea group, hard segement and soft segment. The peak area of urea was relatively small in the DTG curve of the uncured PU adhesive, whereas it noticeably increased in the DTG curve of the PU adhesive cured at 65%RH and 20 °C for 7 days. In the inorganic decomposition region following the organic decomposition region, weight loss occurred between 700 and 800 °C, which corresponds to the decomposition temperature range of $CaCO_3$ (Fig. S5). This residue, including inorganic materials that did not participate in the curing reaction, exhibited a constant weight percentage regardless of the applied relative humidity and curing time (Fig. S4b).

We summarized the variation in the chemical components of the PU adhesive according to the relative humidity and curing time via the DTG curve in the organic decomposition region (Fig. 4c). As the curing progressed, there was little change in the area percentage of the soft segment and plasticizer, which did not initiatively participate in the curing reaction. However, the area percentage of the hard segment and urea changed as a function of the curing time, indicating that the hard

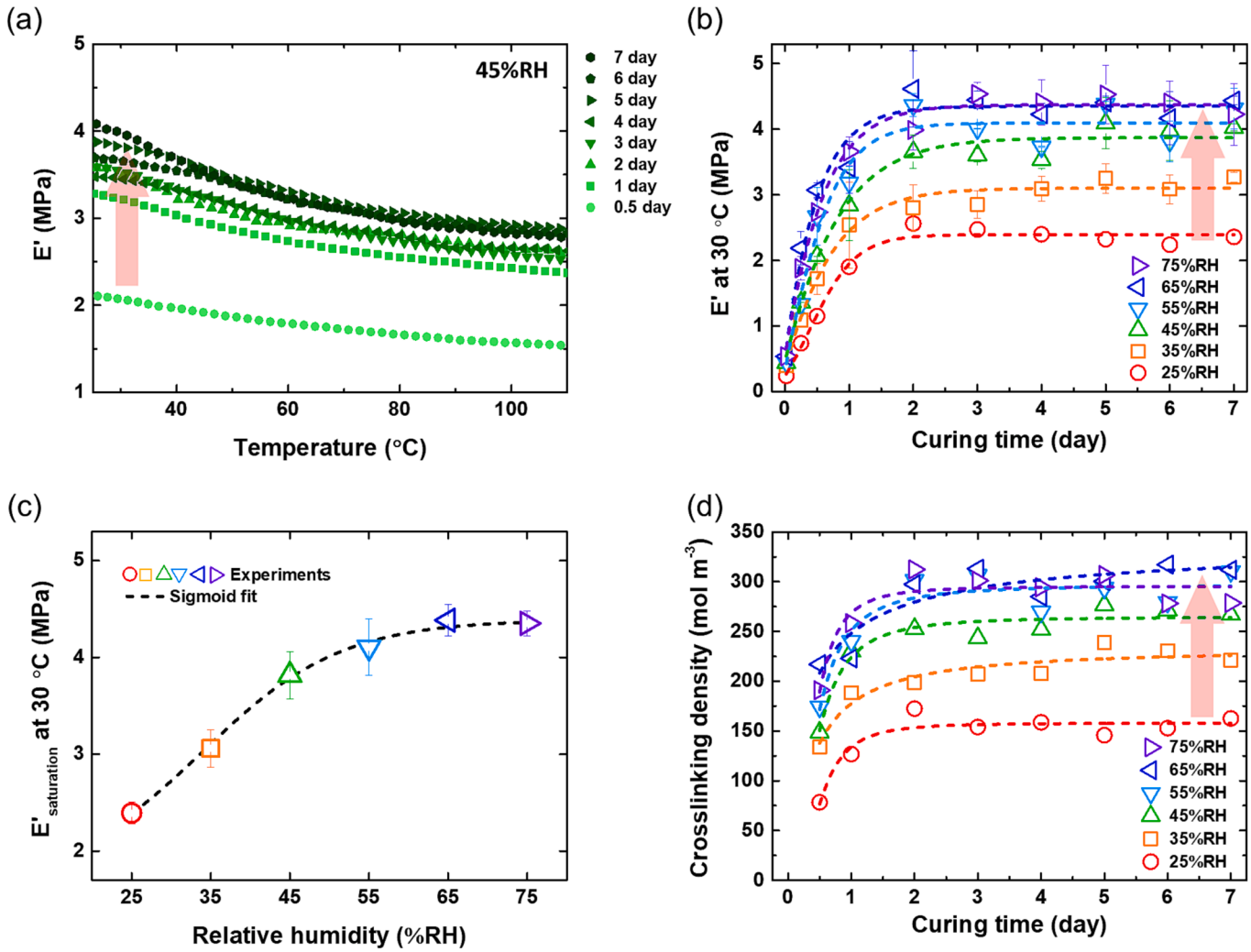

Fig. 5. Mechanical properties of PU adhesive. (a) Storage modulus depending on curing time of adhesive cured at 45%RH and 20 °C, (b) Storage modulus at 30 °C of adhesive cured at different relative humidity and curing time at a constant temperature of 20 °C. The dashed lines indicate sigmoid fit as a function of curing time. (c) Saturated storage modulus at 30 °C of adhesive cured at different relative humidity and a constant temperature of 20 °C for 7 days. (d) Crosslinking density of adhesive cured at different relative humidity and curing time at a constant temperature of 20 °C. The dashed lines indicate fitting via hill equation as a function of curing time.

segment and urea are involved in chemical reactions. The area percentage of urea increased, whereas that of the hard segments decreased as curing progressed. The increase in the area percentage of urea was attributed to the formation of urea bonds from –NCO within the hard segment in response to the surrounding moisture, which resulted in a decrease in the area percentage of the hard segment and a corresponding increase in that of urea. Fig. 4d shows the area percentage of urea within PU adhesive as a function of curing time depending on the different relative humidities at 20 °C. The area percentage of urea exhibited a rapid increase before 2 days under all relative humidity conditions and saturation, followed by a plateau after 2 days. As the relative humidity increased, the area ratio of urea increased more rapidly and significantly, and the saturated area of urea also increased. Under high relative humidity conditions, the formation of urea was promoted by a large amount of moisture, leading to an increase in the final saturated area of urea.

To evaluate the mechanical properties of the PU adhesive in relation to the alterations in its chemical composition, the storage modulus (*E'*) of the PU adhesive cured under diverse curing conditions was measured using DMA with a temperature ramp/controlled force mode (Fig. 5a and S6). As the curing time increased, the storage modulus of the PU adhesive initially increased and then reached saturation, at which point it remained relatively constant. To facilitate a comparison of the storage modulus of the PU adhesive at the same temperature, we evaluated the storage modulus at 30 °C in the DMA data. Fig. 5b shows the storage modulus at 30 °C of the PU adhesive as a function of the curing time, depending on the relative humidity. The storage modulus at 30 °C exhibited a rapid increase before 2 days under all relative humidity conditions and reached saturation after 2 days. The storage modulus at 30 °C was increased more for PU adhesive cured under higher relative humidity conditions during the curing process. Furthermore, the saturated storage modulus at 30 °C increased with higher relative humidity (Fig. 5c). The saturated storage modulus at 30 °C refers to the storage modulus of the adhesive cured at each relative humidity condition for 7 days. The storage modulus at 30 °C of the PU adhesive increased with the curing time and represented a larger saturated storage modulus of the fully cured PU adhesive under high relative humidity. The saturated storage modulus at 30 °C of the PU adhesive cured at 25%RH was 2.39 MPa, while that of the adhesive cured at 75%RH was 4.36 MPa, indicating an increase in the saturated storage modulus with an increase in relative humidity. The adhesives cured at a relative humidity of 65%RH and above exhibited a similar saturated storage modulus. These results suggest that the mechanical properties of the PU adhesive are strongly dependent on the applied relative humidity during the curing process,

beyond which further increases in relative humidity and curing time do not significantly affect the modulus. Furthermore, the storage modulus in the rubbery plateau region was correlated with the number of crosslinks, which was influenced by both the initial humidity conditions and the curing process. The degree of crosslinking is a crucial factor affecting the mechanical strength and viscoelastic properties of PU adhesives. Therefore, we evaluated the crosslinking density (ν_e) of the PU adhesive using Flory's rubber elasticity theory.

$$\nu_e = E'_{high}/3RT_{high}$$

where E'_{high} is the storage modulus in the rubbery plateau region, R is the ideal gas constant, and T_{high} is the absolute temperature in the rubbery plateau region. The crosslinking density of the PU adhesive as a function of curing time exhibited a similar tendency to increase in comparison with the saturated storage modulus at 30 °C (Fig. 5d). The crosslinking density of the PU adhesive cured at 25%RH and 20 °C was 162.7 mol/m^3, whereas that of the adhesive cured at 75%RH and 20 °C was 270 mol/m^3, indicating an increase in crosslinking density with an increase in relative humidity. In an environment with a high relative humidity, the chemical reaction of –NCO occurs rapidly, leading to the formation of more urea bonds and ultimately increasing the degree of curing related to the crosslinking density. Finally, the mechanical properties of these PU adhesives are closely related to the final adhesion performance. The final adhesion performance of PU adhesive was evaluated through a lap shear test of the PU adhesive cured under humidity conditions of 25, 45, 55, and 75%RH for curing time of 7 days. The PU adhesives cured at 25% RH exhibit a lower modulus and the lowest shear strength. In contrast, the PU adhesives cured at 55 and 75 % RH show similar values and reach a state of saturation.

4. Conclusion

We performed a systematic and comprehensive analysis of the effects of relative humidity and curing time on properties of moisture-curable PU adhesives using FT-IR, rheometer, TGA, and DMA analyses. The normalized –NCO peak in the FT-IR spectra decreased exponentially with the curing time and accelerated depending on the relative humidity, eventually disappearing after 5 days at 25%RH and 3 days at 65%RH. An increase in the relative humidity accelerated the formation of urea via a reaction with –NCO by promoting the infiltration of moisture into the PU adhesive. The area percentage of urea in the PU adhesive, as analyzed by TGA, increased and saturated with the increasing curing time and relative humidity. Compared to the curing reaction by chemical variation, the mechanical properties of the PU adhesive exhibited a similar tendency under different curing times and relative humidity conditions. According to the rheological results, the plateau modulus of the initial paste form of the PU adhesives increased with the curing time. However, the increase in the relative humidity and curing time did not lead to a continuous increase in the mechanical properties of the adhesive. The storage modulus of the PU adhesives ceased to increase after 2 days of curing time for all humidity conditions, and the storage modulus at 30 °C of the adhesives cured at 65 and 75% RH were comparable (4.38 and 4.35 MPa, respectively). Finally, the calculation of the crosslinking density demonstrated a correlation between the curing kinetics and the structure–property relationship of the adhesives depending on relative humidity, with the crosslinking densities of the adhesives cured at 25 and 75 %RH being measured at 162 and 270 mol/m^3, respectively. Under high relative humidity conditions, the abundant moisture activated the reaction of –NCO and provided more reaction sites, resulting in the formation of more urea bonds that increased the crosslink density and ultimately enhanced the mechanical properties. Overall, this study provides insights into the effects of the curing conditions on the chemical structure and rheological behavior of PU adhesives, which could be useful for optimizing the curing process and improving the performance of PU adhesives in practical applications.

CRediT authorship contribution statement

Hojun Moon: Formal analysis, Investigation, Visualization, Methodology, Writing – original draft. **Jeong Eun Park:** Formal analysis, Investigation, Visualization, Methodology. **Woongbi Cho:** Formal analysis, Investigation, Visualization, Methodology. **Jaecheol Jeon:** Resources, Conceptualization, Methodology. **Jeong Jae Wie:** Project administration, Conceptualization, Methodology, Supervision, Funding acquisition, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Jeong Jae Wie reports was provided by Hyundai Motor Company. Jeong Jae Wie reports a relationship with Hyundai Motor Company that includes: funding grants].

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112579.

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